

Copper Electroplating

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Students visibly transform a metal object

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 2 — Copper Electroplating

Hook: Students visibly transform a metal object.

Learning outcomes

- Explain electrolysis and distinguish it from a galvanic cell
- Identify cathode and anode in an **electrolytic** cell
- Describe how current controls deposition rate
- Explain why surface preparation and current density affect deposit quality

Session plan (90 min)

Time	Activity	Reference
0–10 min	Show blue CuSO_4 solution / dissolving crystals; preview plating	lecture.md — Hook
10–25 min	Concept: ion deposition at cathode, anode dissolution	lecture.md
25–35 min	Clean cathode object (sand, rinse, no fingerprints)	experiment.md — Prep
35–50 min	Set up plating cell	experiment.md — Setup
50–70 min	Plate 2–10 min; observe every 1–2 min	experiment.md — Plating
70–80 min	Compare variables	experiment.md — Variables
80–90 min	Discussion: surface prep and current density	lecture.md — Wrap-up

Key reactions

Cathode (–): $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$

Anode (+): $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$

Files

- lecture.md — Concept block
- experiment.md — Plating procedure
- materials.md — Checklist
- preparation.md — CuSO_4 solution prep

Status

- [] 0.5 M CuSO_4 mixed and labeled
- [] Plating procedure pilot-tested
- [] Lecture slides/diagrams ready

Session 2 — Preparation

Before Session 2 (instructor)

- [] Prepare **0.5 M CuSO₄** plating bath (e.g. 62 g CuSO₄·5H₂O per 500 mL tap water); label concentration
- [] Pilot plate one nail: record time, current, appearance
- [] Pre-sand a few demo objects
- [] Source resistors if using current limiting

Mixing 0.5 M CuSO₄ (quick recipe)

Batch size	CuSO ₄ ·5H ₂ O (blue crystals)	Water
200 mL	25 g	Fill to 200 mL
500 mL	62 g	Fill to 500 mL
1.0 L	125 g	Fill to 1.0 L

1. Weigh CuSO₄·5H₂O; dissolve in ~¾ of the water; stir until clear blue.
2. Top up to final volume; label "**0.5 M CuSO₄ — Session 2 — do not drink**".
3. Tap water is fine; use distilled only if hard tap water makes the bath milky.

Day-of setup

- [] Goggles and gloves at each station
- [] Aliquot **50–100 mL** CuSO₄ per group jar
- [] One 9 V battery per group (fresh if possible)
- [] Waste bucket labeled "copper waste — do not drain"
- [] Demo cell ready for hook (jar + electrodes + meter)

Open questions / notes

- CuSO₄ concentration mixed (target 0.5 M):
- Resistor value used in pilot:
- Best demo object (nail vs. key):

Session 2 — Materials

Per group

- ☐ 0.5 M CuSO_4 plating solution (50–100 mL)
- ☐ Copper gasket, strip, or wire (anode)
- ☐ Steel nail, screw, coin, washer, or key (cathode)
- ☐ 9 V battery or low-voltage DC supply
- ☐ Alligator clips
- ☐ Multimeter
- ☐ Optional: 100–500 Ω resistor (target ~10–40 mA)
- ☐ Sandpaper or steel wool
- ☐ Jam jar or beaker (~100 mL)
- ☐ Gloves and goggles

Instructor — 0.5 M CuSO_4 stock

Batch	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	Water
200 mL	25 g	to 200 mL
500 mL	62 g	to 500 mL
1 L	125 g	to 1 L

- ☐ Extra copper anode material
- ☐ Waste container for copper-bearing liquids
- ☐ Paper towels for rinsing/drying

Pilot notes

Parameter	Target from pilot
Solution	0.5 M CuSO_4
Solution volume	~50–100 mL
Typical current (with resistor)	___ mA
Good plating time	___ min

Session 2 — Experiment: Copper Electroplating

Instructors: polarity checks, deposit quality, and solution notes — instructor.md.

Copper solution — what to use

You need a blue **copper(II)** plating bath from **copper sulfate**. Mix before class (or shortly before students plate). Do **not** rely on overnight vinegar/copper jars.

Recipe — 0.5 M CuSO_4 (recommended)

Batch size	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (blue crystals)	Water
200 mL (about 2 groups)	25 g	Fill to 200 mL
500 mL	62 g	Fill to 500 mL
1.0 L	125 g	Fill to 1.0 L

Water: tap water is fine. Distilled is slightly better if your tap water is very hard (bath may look milky). Save distilled water for Session 3 silver nitrate.

How to mix:

1. Weigh $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (copper sulfate pentahydrate — the common blue lab salt).
2. Dissolve in $\sim\frac{3}{4}$ of the water; stir until clear blue.
3. Add water to the final volume; label "**0.5 M CuSO_4 — Session 2 — do not drink**".
4. Store covered; goggles/gloves when pouring.

Acceptable range if you must improvise: 0.25–1.0 M. Below ~ 0.25 M plating is slow; much above 1 M can give rough deposits and waste chemical.

Do not add strong acid (industrial baths sometimes use H_2SO_4 — skip that for this age group).

Volume per group today

50–100 mL of 0.5 M CuSO_4 in a jam jar or small beaker (enough to immerse both electrodes).

Prep — Clean the cathode (25–35 min)

Object options

Steel nail, screw, coin, washer, or key (one “good” object + optional unplated control of the same type).

Procedure

1. Sand or steel-wool the surface until **bright metal** shows (no rust, paint, or dull oxide).
2. Rinse under water; pat dry with a paper towel.

3. **Do not touch** the area that will be plated — hold by the head, thread, or edge only. Finger oil ruins adhesion.
4. Optional: photograph or weigh before plating for comparison.

Quality checklist

- [] No visible rust or grease
- [] Uniform bright sanded surface
- [] Rinsed and dry
- [] Placed on a clean paper towel until use

Setup — Plating cell (35–50 min)

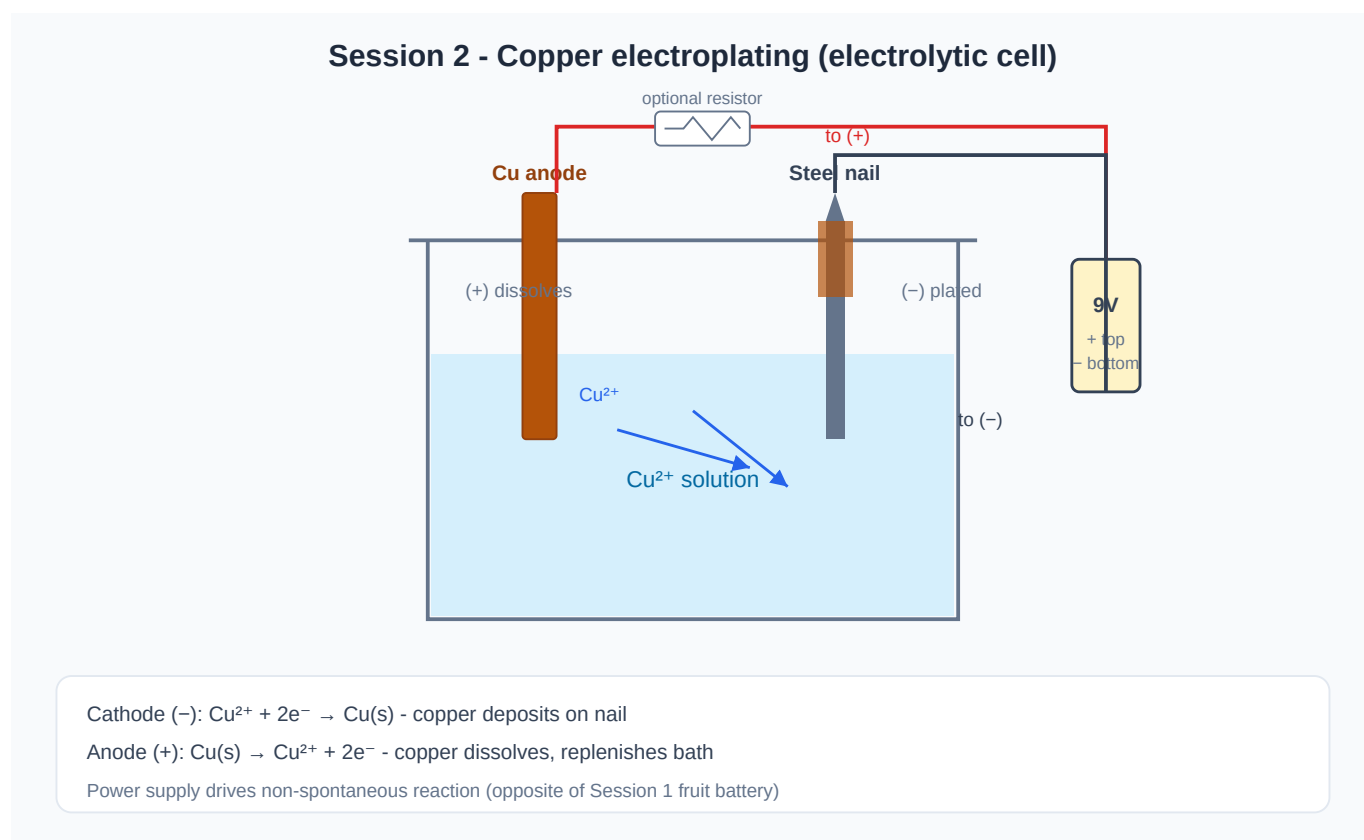


Figure 1 — Electrolytic cell: Cu anode (+) dissolves; Cu^{2+} deposits on the nail cathode (-). Optional resistor limits current.

Materials at the station

- 50–100 mL **0.5 M CuSO_4**
- Copper strip, gasket, or thick wire (**anode**)
- Cleaned object (**cathode**)
- 9 V battery (fresh)
- Alligator clips (at least 2)
- Optional: 100–500 Ω resistor, multimeter on DC mA

Wiring — follow exactly

Part	Connects to
Object to plate (nail, etc.)	Battery negative (-) / black
Copper anode	Battery positive (+) / red
Optional resistor	In series on either lead (limits current)
Optional ammeter	In series on DC mA range

Memory aid: *The object that should gain copper goes to the minus sign.*

Procedure

1. Pour **50–100 mL** CuSO_4 solution into the jar.
2. Clip the **copper anode** to battery **(+)**; clip the **object** to battery **(-)**.
3. If using a resistor: put it in series so current must pass through it.
4. If using a multimeter for current: set to DC mA; wire it **in series** (not across the battery like a voltmeter).
5. Lower both electrodes into the solution so the plating area is submerged.
6. Keep electrodes **2–4 cm apart**; they must **not touch** (touching = short circuit).
7. Confirm bubbles or a slow color change starting within ~30–60 s. Read initial current if possible (**target ~10–40 mA** with a resistor; without a resistor current may be higher).

Pre-plating record

Item	Value
Solution (0.5 M CuSO_4)	
Approximate concentration if known	
Solution volume (mL)	
Solution color	
Anode material	
Cathode object	
Electrode spacing (cm)	
Initial current (mA)	
Resistor used? (Y/N, Ω)	

Plating run (50–70 min)

1. Start the timer when the circuit is connected and current is flowing.
2. Every **1–2 minutes**, look at the object: color, shine, roughness, bubbles.
3. Typical good run: **3–8 minutes**. Stop sooner if the deposit turns black or powdery.

4. Disconnect the battery **before** removing the object.
5. Rinse gently under water; pat dry. Do not scrape the new copper.
6. Compare to an unplated control of the same object type.

Observation log

Time (min)	Appearance	Current (mA)	Notes
0			
2			
4			
6			
8			
Final			

What “good” looks like

Appearance	Meaning	Action
Bright salmon / copper color	Good plating	Continue or stop at planned time
Dark brown / black powder	Current too high	Stop; add resistor; re-sand; restart shorter
Patchy	Dirty surface or poor immersion	Re-clean; ensure full immersion
Heavy bubbles on object	Competing H ₂ evolution	Lower current / increase spacing

Variable comparisons (70–80 min)

Groups test **one** variable (share results):

Variable	Options	What to compare
Plating time	2 vs. 8 min	Thickness, color
Current	With vs. without 220–470 Ω resistor	Smoothness
Surface prep	Clean vs. finger-greased	Adhesion, coverage
Concentration	~0.25 M vs. 0.5 M (if you dilute a portion)	Speed, color quality
Distance	~1 cm vs. ~4 cm electrode gap	Uniformity, current

Results table (class compilation)

Group	Variable	Result	Best practice conclusion

Troubleshooting

Problem	Cause	Fix
No plating	Reversed polarity	Object must be on (-)
No plating	Solution too weak / wrong bath	Remake 0.5 M CuSO ₄ ; check polarity
Black/powdery Cu	Current too high	Add 100–500 Ω resistor; increase distance
Patchy deposit	Dirty / oily surface	Re-sand, rinse, no fingerprints
Anode not dissolving	Wrong anode (not copper)	Use copper strip/wire
Battery gets hot	Short circuit	Separate electrodes; check clips
Very slow plating	Low [Cu ²⁺] or bad contacts	Remake / concentrate bath; clean clips

Safety during experiment

- Goggles on when working with solutions; gloves recommended
 - Do not ingest solutions; no mouth pipetting
 - Avoid short-circuiting the 9 V battery
 - Collect all copper-bearing liquid in the labeled waste container — **not** down the drain
 - Wash hands after the lab
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Experiment status

- ☐ 0.5 M CuSO₄ mixed and labeled
- ☐ Optimal plating time determined in pilot (record: ____ min)
- ☐ Resistor value chosen for ~10–40 mA (record: ____ Ω)
- ☐ Before/after photos for lecture use
- ☐ Variable comparison sheet printed for groups

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

The three rules that matter most

1. **Salt belongs in exactly one place this week.** NaCl is used only for the Session 5 corrosion jars, where nothing is driven hard enough to make chlorine. It is **forbidden** in every electrolysis bath and in the derusting cells, where chloride is oxidized at the anode to **chlorine gas**. This is a competing half-reaction, not an arbitrary rule.
2. **Any cell left running unattended must be open and ventilated.** Electrolysis and derusting cells evolve hydrogen continuously. An overnight derusting cell can generate a couple of litres of it. Open jars, ventilated room, no ignition sources, nothing sealed. Ever.
3. **Disconnect power before touching anything wet.** Every time, as a ritual, especially at the Session 5 reveal.

General rules

1. **No eating experimental materials.** Fruit, potatoes and liquids used in experiments are no longer food, and this includes the ones that started out edible.
2. **Ammeters go in series.** A meter on a current range clipped across a supply is a short circuit and will blow its fuse at best.
3. **Avoid short circuits generally.** Batteries heat up fast when shorted.
4. **Collect metal-salt waste.** Copper and silver solutions are not poured down the drain.
5. **Water choice:** tap water is fine for most baths. **Distilled is required for the Session 3 silver bath**, so chloride does not precipitate silver, and is worth having for the first rung of the Session 4 conductivity ladder.
6. **Goggles for all wet work.** Gloves additionally for silver nitrate.
7. **Rusty hardware is sharp** and often outdoors-dirty. Forceps or gloves; no bare-handed scrubbing.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails and wires	Supervise handling; wash hands after handling metals
2	Copper sulfate solution	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves and goggles; stains skin and clothing near-permanently; small volumes (30–40 mL); collect all waste; no ammonia, ever
4	Gas evolution; hydrogen ignition	No NaCl; ammeter in series; U-tube side arms open; balloon and flame instructor only
5	Corrosion demos; overnight derusting	Washing soda only, never NaCl; disconnect before handling; jars open and ventilated overnight

Silver nitrate and ammonia — Session 3

Never add ammonia, or any ammonia-containing cleaner, to a silver nitrate solution. Ammoniacal silver solutions (Tollens' reagent) deposit **silver nitride and related compounds** on standing, which are friction- and shock-sensitive explosives. The hazard is not the fresh solution — it is any residue left in a bottle, on a stirrer, or around a cap.

This matters because the "silver mirror" recipes students will find online use exactly this chemistry. Say so out loud during the Session 3 safety brief, before someone tries it at home.

It also does not even work for our purpose: ammonia complexes copper about as strongly as it complexes silver, so it does nothing to suppress the displacement reaction we are trying to control.

Hydrogen — Session 4, instructor only

The balloon demo is optional. The chemistry is fully demonstrated by the splint tests without any of this risk, so skip the balloon entirely if the venue forbids open flame or if you are at all unsure.

If you do run it:

- Feed **cathode (H₂) gas only** into the balloon. The anode vents separately to the room.
- **Never a deliberate H₂/O₂ mixture.** Pure hydrogen in air *deflagrates* — it burns at the surface and gives a bark. A stoichiometric mixture **detonates**, which is a different event entirely and not a classroom demonstration.
- **Never a rigid or sealed container.** A balloon tears; glass becomes shrapnel.
- Modest inflation only.
- Tie off, then carry it **well away** from the electrolysis cell, which is still making hydrogen.
- Long lighter at arm's length; area cleared; extinguisher present and its location known.
- Warn the room. The bang is loud, sudden, and some students find it distressing.

Splint tests — the low-risk alternative

Gas	Test	Result
H ₂	Lit splint at the mouth of the tube	Squeaky pop
O ₂	Glowing (not flaming) splint inserted	Ember relights

A few millilitres of gas in a small test tube is plenty. Keep hydrogen tubes **mouth-down** until the moment of testing. TA-supervised; students may hold the tube, an adult holds the flame.

Waste

Waste	Disposal
Copper sulfate solution	Labeled waste container; not down the drain
Silver nitrate solution and rinses	Labeled waste container; not down the drain. Drop a coil of bare copper wire in — it strips the silver back out as metal for recovery
Sodium sulfate electrolyte	Drain is acceptable; dilute well

Waste	Disposal
Washing soda derusting sludge	Let solids settle, bag the sludge, dilute the liquid to drain
Saltwater corrosion jars	Drain; dispose of rusty metal in solid waste